4 Planes and Surfaces

4.1 Determining a Plane

A plane $\mathcal{P}$ in $\mathbb{R}^3$ can be uniquely determined by any of the following geometric data:

- A point Q on the plane and a normal vector $\vec{n} \perp \mathcal{P}$.
- Three non-collinear points.
- A general linear equation $ax + by + cz + d = 0$.
- Two intersecting lines (or two non-parallel direction vectors).
- Two parallel distinct lines.
- A line $\mathcal{L}$ and a point $Q \notin \mathcal{L}$.

4.2 Mathematical Formulations of a Plane

Case 1: Plane passing through the origin $O(0, 0, 0)$

A vector $\vec{v} = \langle x, y, z \rangle$ lies in $\mathcal{P} \iff \vec{v} \perp \vec{n}$, where $\vec{n} = \langle a, b, c \rangle$:

$$\vec{v} \in \mathcal{P} \iff \vec{v} \cdot \vec{n} = 0 \iff ax + by + cz = 0$$

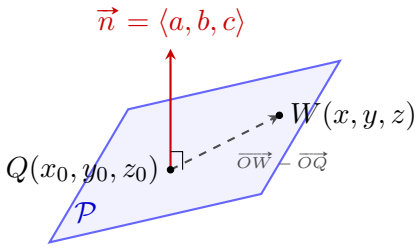
In set notation:

$$\mathcal{P} = \{\langle x, y, z \rangle \mid ax + by + cz = 0\}$$

Case 2: Plane passing through a point $Q(x_0, y_0, z_0)$

Let Q be a known point on $\mathcal{P}$ with position vector $\overrightarrow{OQ} = \langle x_0, y_0, z_0 \rangle$. Any point $W(x, y, z)$ on the plane satisfies:

$$(\overrightarrow{OW} - \overrightarrow{OQ}) \perp \vec{n} \iff (\overrightarrow{OW} - \overrightarrow{OQ}) \cdot \vec{n} = 0$$



Expanding this dot product yields the point-normal form:

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$$

or equivalently:

$$ax + by + cz = ax_0 + by_0 + cz_0$$

We can always scale the coefficients a, b, c, d in $ax + by + cz = d$ by $\frac{1}{\|\vec{n}\|}$ to obtain a unit normal vector $\hat{n} = \frac{\langle a,b,c \rangle}{\sqrt{a^2+b^2+c^2}}$.

4.3 Properties of Normal Vectors

For any set of vectors $\{\vec{u}, \vec{v}, \vec{w}\}$ where $\vec{v} \nparallel \vec{w}$, the following statements are equivalent:

- $\vec{u} \perp \vec{v}$ and $\vec{u} \perp \vec{w}$.
- $\vec{u} \parallel (\vec{v} \times \vec{w})$.
- $\vec{u} \perp \mathcal{P}$, where $\mathcal{P} = \text{Span}\{\vec{v}, \vec{w}\}$.

Problem 4.1: Finding Normal Vectors

Find a normal vector $\vec{n}$ to the plane $\mathcal{P}$ given by the equation $(x - 1) - (y - 1) = 0$.

Solution 4.1

The equation simplifies to $x - y = 0$. We can extract or compute the normal vector in two ways:

Method 1 (Direct Inspection): From the general form $ax + by + cz = d$, the coefficients of x, y, z directly give the normal vector:

$$\vec{n} = \langle 1, -1, 0 \rangle$$

Method 2 (Cross Product of Parallel Plane Vectors): Two vectors parallel to the plane $\mathcal{P}$ are $\vec{v_1} = \langle 0, 0, 1 \rangle$ and $\vec{v_2} = \langle 1, 1, 0 \rangle$. Taking their cross product gives:

$$\vec{n} = \vec{v_1} \times \vec{v_2} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{vmatrix} = \vec{i}\,(0-1) - \vec{j}\,(0-1) + \vec{k}\,(0-0) = \langle -1, 1, 0 \rangle$$

Note that $\langle -1, 1, 0 \rangle = -\langle 1, -1, 0 \rangle$, which represents the same normal line direction.

Problem 4.2: Finding a Plane From Three Points

Find the equation of the plane containing the points $P(1, 0, 1)$, $Q(0, 2, 0)$, and $R(2, 1, 0)$.

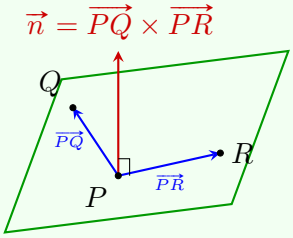
Solution 4.2

Step 1: Construct two displacement vectors in the plane

$$\overrightarrow{PQ} = \langle 0-1, 2-0, 0-1\rangle = \langle -1, 2, -1\rangle$$
$$\overrightarrow{PR} = \langle 2-1, 1-0, 0-1\rangle = \langle 1, 1, -1\rangle$$

Step 2: Compute the normal vector via cross product

$$\vec{n} = \overrightarrow{PQ} \times \overrightarrow{PR} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -1 & 2 & -1 \\ 1 & 1 & -1 \end{vmatrix} = \langle -1, -2, -3\rangle$$



Step 3: Write the plane equation using point $P(1, 0, 1)$

$$-1(x-1) - 2(y-0) - 3(z-1) = 0$$

Simplifying yields:

$$x + 2y + 3z = 4$$

5 Spheres and Planes

5.1 Determining a Sphere

A sphere in $\mathbb{R}^3$ can be uniquely determined by any of the following sets of data:

- A center point $(a, b, c) \in \mathbb{R}^3$ and a radius $R \in \mathbb{R}^{>0}$.
- Two points representing the endpoints of a diameter.
- Four non-coplanar points.

- A plane tangent to the sphere together with three points lying on the sphere.

5.2 Plane-Sphere Intersections

Example 5.1.

Consider the intersection of the sphere $x^2 + y^2 + z^2 = 4$ with the line $\ell(t) = (t, 0, 1)$ passing through $(1, 0, 1)$ and $(-1, 0, 1)$.

Substituting $\ell(t)$ into the sphere equation yields:

$$\begin{aligned} t^2 + 0^2 + 1^2 &= 4 \\ t^2 + 1 &= 4 \\ t^2 = 3 &\implies t = \pm\sqrt{3} \end{aligned}$$

Thus, the line intersects the sphere at $\ell(\sqrt{3}) = (\sqrt{3}, 0, 1)$ and $\ell(-\sqrt{3}) = (-\sqrt{3}, 0, 1)$.

5.3 Traces of Surfaces

Definition 5.1: Traces

A **trace** of a surface is the intersection curve obtained by intersecting the surface with a parallel coordinate plane of the form $x = k$, $y = k$, or $z = k$.

Example 5.2 (Traces of a Sphere)**.**

Consider the sphere given by:

$$(x - a)^2 + (y - b)^2 + (z - c)^2 = R^2$$

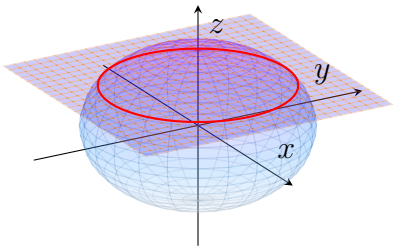
Taking the trace with the plane $z = k$ yields the equation:

$$(x - a)^2 + (y - b)^2 = R^2 - (k - c)^2$$

Depending on the plane distance k, we have three distinct cases:

- **Case 1 ($|k - c| = R$):** The trace is a single point at (a, b, k).

- **Case 2 ($|k-c| < R$):** The trace is a circle of radius $\sqrt{R^2-(k-c)^2}$ centered at (a, b, k).
- **Case 3 ($|k-c| > R$):** The trace is empty (no real intersection).



Example 5.3.

Find the trace of the plane $\mathcal{P} = \{(x, y, z) \mid -2x + 6y - 5z = 4\}$ with the plane $x = -1$:

$$\begin{aligned} -2(-1) + 6y - 5z &= 4 \\ 2 + 6y - 5z &= 4 \\ 6y - 5z = 2 \implies z &= \frac{6}{5}y - \frac{2}{5} \end{aligned}$$

The trace is a straight line in the plane $x = -1$.

5.4 Quadric Surfaces

A general **quadric surface** in $\mathbb{R}^3$ is given by the second-degree equation:

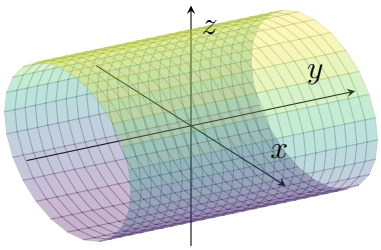
$$Ax^2 + By^2 + Cz^2 + Dxy + Eyz + Fzx + ax + by + cz + d = 0$$

When $D = E = F = 0$, the surface is in **standard position** (axes aligned with coordinate axes).

Example 5.4.

The surface $x^2 + z^2 = 1$ is a circular cylinder aligned along the y-axis with radius 1.

- $z = 0$ trace: $x^2 = 1 \implies x = \pm 1$ (parallel lines along y).
- $x = 0$ trace: $z^2 = 1 \implies z = \pm 1$ (parallel lines along y).

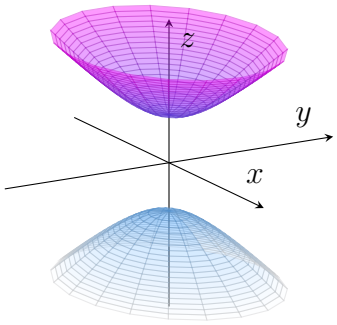


Example 5.5.

Consider the surface given by $x^2 + 4y^2 - z^2 = -1$.

Traces:

- $x = 0$ **trace:** $4y^2 - z^2 = -1 \implies z = \pm\sqrt{4y^2 + 1}$. As $|y| \to \infty$, $z \approx \pm 2y$ (Hyperbola).
- $y = 0$ **trace:** $x^2 - z^2 = -1 \implies z = \pm\sqrt{x^2 + 1}$. As $|x| \to \infty$, $z \approx \pm x$ (Hyperbola).
- $z = k$ **trace:** $x^2 + 4y^2 = k^2 - 1$:
  - $|k| = 1$: Single point at $(0, 0, \pm 1)$.
  - $|k| < 1$: No trace (empty set).
  - $|k| > 1$: Ellipse given by $x^2 + 4y^2 = k^2 - 1$.



Why do cross-sectional ellipses fall on the surface?

For any fixed height $z = k$ with $|k| > 1$, substituting $z = k$ fixes the value $k^2 - 1 > 0$ on the right side of $x^2 + 4y^2 = k^2 - 1$. This traces a locus of points (x, y, k) whose horizontal distance components continuously satisfy the surface equation for that slice.

5.5 Hyperbolic Paraboloid and Its Traces

Definition 5.2: Quadric Surface

A quadric surface is defined by a quadratic equation in three variables x, y, and z:

$$Ax^2 + By^2 + Cz^2 + Dxy + Eyz + Fxz + ax + by + cz + d = 0$$

, where $A, B, C, D, E, F, a, b, c, d$ are constants, and at least one of A, B, C, D, E, F is non-zero.

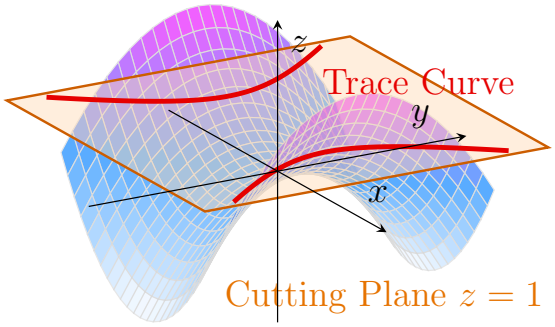
Definition 5.3: Trace

A trace is the intersection of a quadric surface with a plane, usually by taking slices parallel to the coordinate planes.

Note that when our surface can be written by $z = f(x, y)$, the traces form level sets of f.

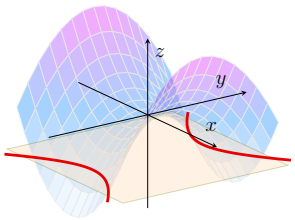
Example 5.6.

Consider the quadratic equation $z = x^2 - y^2$. Now take the traces in $z = -1$, $z = 0$, and $z = 1$. Make a sketch of the traces in $x = -1$, $x = 0$, and $x = 1$. In general, the traces in $x = k$ when k is a constant are hypobolas with equations $z = k^2 - y^2$.

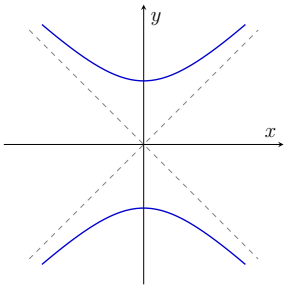


Now take the traces in $z = -1$, $z = 0$, and $z = 1$.

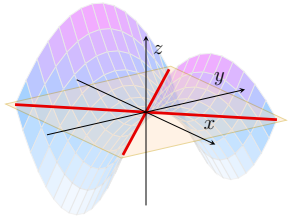
Trace at $z = -1$



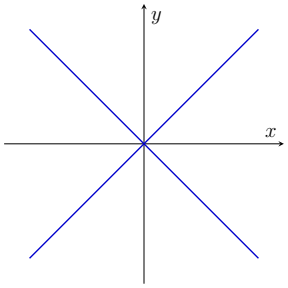
$y^2 - x^2 = 1$



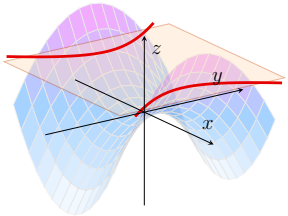
Trace at $z = 0$



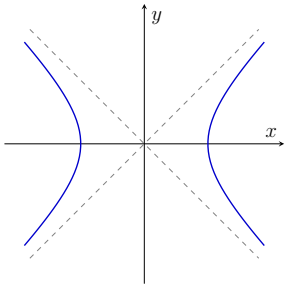
$y = \pm x$



Trace at $z = 1$

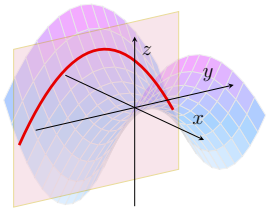


$x^2 - y^2 = 1$

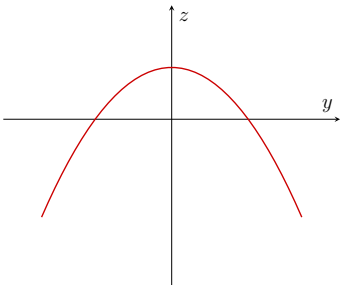


Make a sketch of the traces in $x = -1$, $x = 0$, and $x = 1$. Slicing the surface with vertical planes $x = k$ parallel to the yz-plane yields parabolic trace curves.

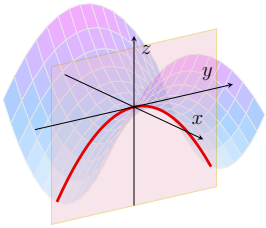
Trace at $x = -1$



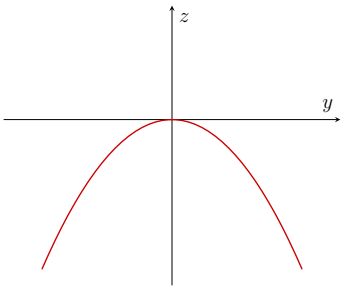
$z = 1 - y^2$



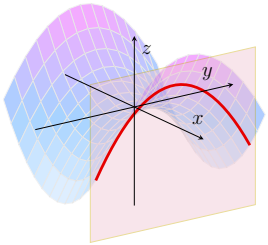
Trace at $x = 0$



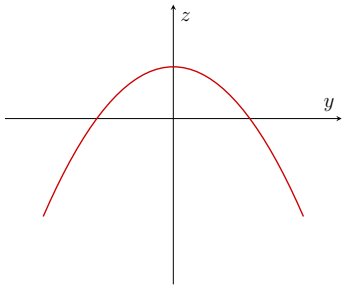
$z = -y^2$



Trace at $x = 1$



$z = 1 - y^2$



Ellipsoids are another family of quadric surfaces. The general equation of an ellipsoid is given by:

$$\left(\frac{x}{a}\right)^2 + \left(\frac{y}{b}\right)^2 + \left(\frac{z}{c}\right)^2 = 1$$